

Emerging Technologies and Their Impact on Disability

Paul H. Wise

Summary

Technological innovation is transforming the prevalence and functional impact of child disability, the scale of social disparities in child disability, and perhaps the essential meaning of disability in an increasingly technology-dominated world. In this article, Paul Wise investigates several specific facets of this transformation. He begins by showing how technological change influences the definition of disability, noting that all technology attempts to address some deficiency in human capacity or in the human condition.

Wise then looks at the impact of technology on childhood disabilities. Technical improvements in the physical environment, such as better housing, safer roads, and poison-prevention packaging, have significantly reduced childhood injury and disability. Other technological breakthroughs, such as those that identify genetic disorders that may lead to pregnancy termination, raise difficult moral and ethical issues. Technologies that identify potential health risks are also problematic in the absence of any efficient treatment.

Wise stresses the imbalance in the existing health care delivery system, which is geared toward treating childhood physical illnesses that are declining in prevalence at a time when mental and emotional conditions, many of which are not yet well understood, are on the rise. This mismatch, Wise says, poses complex challenges to caring for disabled children, particularly in providing them with highly coordinated and integrated systems of care.

Technology can also widen social disparities in health care for people, including children with disabilities. As Wise observes, efficacy—the ability of a technology to change health outcomes—is key to understanding the relationship of technology to social disparities. As technological innovation enhances efficacy, access to that technology becomes more important. Health outcomes may improve for those who can afford the technology, for example, but not for others. Hence, as efficacy grows, so too does the burden on society to provide access to technology equitably to all those in need. Without such access, technological innovation will likely expand disparities in child outcomes rather than reduce them.

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Technology has long been recognized as a potential way to help ensure that children with disabilities will have optimal opportunity for a long, healthy, and socially engaged life. Traditionally, technology and other interventions designed for children with disabilities were focused on strategies aimed at correcting a child's specific impairment or deficit. New scholarship and decades of disability advocacy have expanded this purview to include a wide variety of environmental and societal factors that are now recognized to be essential in optimizing health, development, and social engagement for children with disabilities. This more comprehensive understanding emphasizes the dynamic interaction between the physical environment and the technological and social forces that can reshape it.

Today the prevention and treatment of disability in childhood are being recast by unprecedented technological innovation. In essence, the nature and cadence of this innovation are transforming the prevalence and functional impact of child disability, the scale of social disparities in child disability, and perhaps the essential meaning of disability in an increasingly technology-dominated world. This article investigates several specific facets of this transformation: the influence of technological change on the definition of disability, the impact of preventive and therapeutic interventions on disabilities in childhood, and the ability of the current delivery system to afford access to emerging technologies designed to prevent and reduce the impact of disabling conditions in children. The article also discusses the interaction of technical innovation and the social determinants of health in shaping patterns of childhood disability as well as the interaction between the diffusion of science

and technology design and disparities in child health. Understanding these issues and interactions is helpful in designing the health care delivery systems, programs, and public policies that will ultimately prove most effective in addressing childhood disabilities in the years to come.

Defining Disability and Assistive Technology

The definition of technology used in this discussion is comprehensive in nature and refers to the application of scientific knowledge for practical, applied purposes, here directed toward improving health and well-being. The definition of disability has undergone dramatic evolution over the years, conforming to evolving analytical frameworks and societal perceptions. For the purposes of this discussion, I use the definition of disability proposed by Neal Halfon and his colleagues in their article in this volume:

A disability is an environmentally contextualized health-related limitation in a child's existing or emergent capacity to perform developmentally appropriate activities and participate, as desired, in society.¹

In relation to this definition, technology can refer to both preventive and therapeutic interventions and can take on a variety of forms, including vaccines, other pharmaceuticals, engineering, or alterations to the physical or social environment. A primary objective is the maximization of a child's ability to function independently, which is in many ways determined by the ability to perform essential daily tasks, including those involving hygiene, mobility, and social interaction.² Another central objective is the minimization of the impact that the child's disability has on caregivers, both in their

provision of direct assistance and more generally as part of day-to-day family life.³

A careful examination of the relationship between disability and technology, however, raises important questions related to the definition and societal meaning of disability in the face of rapidly changing technological capabilities. First, a changing technological environment can dramatically alter the functional impact of any given disability. For example, the development of the telephone greatly enhanced communication in general society. At the same time, the central importance of aural communication in a telephone-dominated society made deafness an increasingly debilitating disability. Similarly, the emergence of a computer-dominated society and its text-based reliance on e-mail and cell phone texting has placed new burdens on the blind. Second, the dynamic interaction between disabilities and technology development underscores the rather arbitrary nature of disability definitions. Virtually all technologies attempt to address some deficiency in human capacity or in the human condition. Automobiles address human inability to move quickly over long distances; telephones address their inability to communicate with their voice over long distances; typewriters and their successors compensate for poor and slow penmanship. At some level, therefore, the definition of disability and the role of technology reflect both the prevalence of a lack of a particular capability and the social response to it. The interactions between disability and technology are, therefore, intensely dynamic and generally evade static categorization or definitions. Indeed, these interactions are undergoing such rapid evolution that they have generated a proliferation of philosophical challenges that have transcended the meaning of disability to seek the meaning of being human.

The Impact of Preventive and Therapeutic Technologies on Childhood Disabilities

Technological innovation has dramatically altered the landscape of both preventive and therapeutic approaches to childhood disability. Advanced preventive strategies reflect new capacities to reduce the occurrence of a disabling condition. The development of a broad array of new vaccines has helped prevent a variety of infectious diseases, such as meningitis, which in turn can result in serious disabling sequelae. Technologies have also played an important role in the early diagnosis of potentially disabling conditions, such as phenylketonuria and other genetic disorders; early diagnosis can permit the early implementation of preventive interventions, including dietary alteration. Rapid progress in therapeutic interventions has also in many instances reduced the impact of disability on daily functioning and social engagement.

Preventive Technologies

Technical innovation has had a dramatic impact on a central arena of primary disability prevention: the reduction of serious, disabling injuries in children. The importance of this preventive domain stems not only from the significant contribution that injuries make to disabling conditions in childhood but also from the strong evidence that injuries are highly preventable. Technical improvements in the physical environment of children, including housing, automobile travel, pedestrian and water safety, medication and poison packaging, and playground design, have led to significant reductions in injury-related mortality and disability in children.⁴ These examples also highlight the interactions between the legal environment, which has mandated safety improvements, and the development of technologies to meet these standards.

Many of these technical improvements benefit all their users because they are based on general design enhancements such as safer roads and automobiles. Other interventions that prevent injuries to children depend for their effectiveness on financial access (buying a child car seat, for example), parental behaviors (using a child car seat or a child-protective car window lock), or both. Many of these interventions are mandated by law, but persistent social disparities characterize their actual use and, consequently, patterns of serious childhood injury.⁵ Technical innovation has also revolutionized the identification of children at risk for childhood disabilities. In large measure, this technology has taken the form of screening initiatives designed to identify and respond to genetic or other indicators of disability risk before a child is conceived, during gestation, or shortly after birth. Genetic screening of prospective parents has dramatically reduced the prevalence of certain relatively rare conditions, such as Tay-Sachs disease.⁶ The ability to identify risk-associated genetic profiles or biomarkers in pregnancy, however, has proven to be the most active, and a highly controversial, arena of technical innovation in disability prevention. The ability to identify the presence of genetic disorders such as trisomy 21 and cystic fibrosis in the fetus, as well as biomarkers or anatomical indicators of disabling pediatric conditions, has traditionally been linked to pregnancy termination, raising difficult ethical and moral questions. Technology, however, is also developing new prenatal interventions, including fetal surgery, that may be able to correct conditions likely to produce disabling damage either later in pregnancy or subsequent to birth.

The use of prenatal diagnostic technology is also characterized by significant social disparities, particularly when complex

medical procedures or delivery infrastructures are required.⁷ Yet, the continued link of prenatal diagnosis to pregnancy termination has made the disparate use of prenatal screening hard to interpret. Social differences in the acceptability of abortion and in access to abortion could also be contributing to observed disparities in the use of prenatal diagnostic procedures.⁸ Differences in access to and use of abortion are likely to be important in explaining disparities in the number of children born with fetal conditions that can be identified through widely available screening approaches, such as ultrasound.

Preventive strategies have also been directed at identifying disabling conditions in newborn infants. These strategies have traditionally involved screening programs designed to identify affected children early enough to implement preventive interventions. This approach, in turn, has usually required that the condition be present but not clinically recognizable at birth and that the condition be amenable to early intervention. Newborn screening programs were initiated in the 1960s to identify children with phenylketonuria. This genetic disorder can cause cognitive impairment that can be prevented by the early initiation of a special, phenylalanine-poor diet. Over the subsequent decades, state health agencies have implemented universal newborn screening programs, and tests for a number of other conditions, including sickle cell disease and cystic fibrosis, have been added to screening protocols.⁹

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health conditions, including cardiovascular disorders and Alzheimer's disease, has been directly marketed to consumers even though the strength of these associations may be weak. Therefore, while the technical ability to identify risk continues to grow, so too does the challenge of making sense of this knowledge and using it to craft an efficient, effective, and humane response.¹⁰

Therapeutic Technologies

In general, children with disabilities rely more heavily than other children on technical interventions, including medications, specialized medical and educational services, and a variety of assistive devices. The term "assistive technology device" was initially documented in federal legislation in the United States as part of the Technology-Related Assistance for Individuals with Disabilities Act of 1988. The proposed definition was "any item, piece of equipment or product system—whether acquired commercially, modified, or customized—that is used to increase, maintain, or improve functional capabilities of individuals with disabilities." Despite changes in the supporting legislation in 1994 and 1998, this definition has remained largely intact and in widespread use.

Between 9 and 15 percent of children in the United States need or use a prescription medication for an ongoing health condition. Indeed, a requirement for prescription medication is the most commonly met criterion for designating a child as having a special health care need.¹¹

One study found that approximately 36 percent of children with special health care needs had a reported need for eyeglasses or vision care; 7 percent required hearing aids or care; and 5 percent required mobility aids or devices.¹² Several national studies reported that approximately one in seven children with special health care needs had at least one unmet need for medical, dental, mental, or other health service.¹³ Approximately half of all children with special health care needs require assistive or medical devices, with 12 percent requiring communication, mobility, or hearing devices. Fourteen percent of these children were found to have unmet assistive technology needs.¹⁴

Studies of specific conditions, particularly cerebral palsy, have documented the importance of technologies designed to improve the functional abilities of children with cognitive and motor disorders¹⁵ and to enhance education, social functioning, and lifelong learning among children and youth with intellectual disabilities.¹⁶ A study of disabled children in an urban area of Finland found that 77 percent of surveyed families benefited from assistive devices for feeding, dressing, and hygiene, particularly if the child had significant motor but mild cognitive disabilities.¹⁷

While access to therapeutic and assistive technology is important, evaluations of the effects of these technologies on child functioning and quality of life remains spotty.

Using classification domains outlined in the World Health Organization's International Classification of Functioning, Disability, and Health,¹⁸ a recent systematic review¹⁹ found that most studies of functioning and quality of life were concerned with technologies designed to enhance communication through new, computer-based modalities²⁰ and to improve mobility through advanced engineering and robotics.²¹ One striking finding was the paucity of assessments of the impact of assistive technology on caregivers and on the children's families.

Overall, this literature suggests that therapeutic and assistive technologies can improve daily functioning primarily through enhancing activity levels and participation in normal activities. However, these published studies reflect a wide variation in the conditions and types of assistive technologies examined, methodological rigor, analytical strategies, and child and family outcomes. Moreover, there may be a significant bias against reporting negative findings because many of these studies evaluated novel or prototypical devices or programs.

The intense interaction of impairment and social context is reflected in significant regional variation in the ways that technology can affect activity levels, participation in normal activities, and the quality of life among disabled children. A recent study of children with cerebral palsy in six European countries documented considerable variation across the eight study regions in the intensity and nature of a child's participation in daily activities and in children's social roles.²² Another far-reaching study of childhood disability in Europe strongly suggested that a substantial portion of this variation resulted from variation in state policies addressing the use of assistive technologies among children

with disabilities.²³ For example, in Denmark, the country with the highest reported levels of participation in daily activities, advocates for disabled children worked closely with the government to facilitate the provision of assistive technologies and the participation of disabled children in a variety of school and after-school activities.²⁴

The Impact of New Technologies on the Prevalence of Childhood Disability

Despite a strong record of successful preventive and therapeutic strategies, there remains a powerful undercurrent of concern that technical innovation has also increased the prevalence of disabilities in childhood. The first mechanism by which technical innovation could be increasing the number of children with disabilities is by shifting mortality into chronic morbidity. While this shift can occur for a variety of serious conditions affecting young children, the decline in neonatal mortality among high-risk newborns, particularly those born prematurely, is of special concern.²⁵ The well-documented reductions in neonatal mortality over the past several decades are attributable primarily to dramatic improvements in the survival of extremely premature infants. While surviving, however, many of these infants go on to suffer from a variety of medical and developmental sequelae, including lung and eye disease, neurologic deficits, and learning disorders.²⁶ Still, the increase in the survival of premature infants is not large enough to account for a major portion of the observed increases in rates of disability.

The improvements in the care of high-risk newborns that have shifted mortality to morbidity in extremely premature neonates have also reduced long-term morbidity in somewhat less premature newborns who previously would have experienced high rates of serious illness and disability.²⁷ The

year-to-year reductions in morbidity lag somewhat behind those in mortality, however, a trend that indicates a rising prevalence of serious disabling conditions emerging from the newborn period. Nonetheless, the impact of technical innovation on both the reduction and the generation of disabling childhood conditions is exceedingly dynamic and should be examined with an informed, analytical eye.

A second, more direct mechanism by which technical interventions could increase the prevalence of serious childhood disabilities is through increasing the number of infants born with a high risk for disabilities. A variety of medications, such as anticonvulsants and retinoids, have been associated with congenital anomalies and other childhood disorders when taken during the prenatal period.²⁸ Assisted reproductive technology, including in vitro fertilization, has been associated with premature birth and low birth weight, in part because of its tendency to result in multiple gestations (twins, triplets, quadruplets). In fact, a significant portion of the increase in the prematurity rate in the United States over the past two decades is estimated to be the result of the growing use of assisted reproductive technology.²⁹

Beyond these discrete, well-documented examples, broader misgivings regarding the potential health impacts of new technical interventions can emerge even for highly efficacious interventions, such as immunizations, when the etiology of a major disabling condition, such as autism or asthma, is poorly understood.³⁰ Although there remains no evidence that immunizations heighten the risk of autism or asthma, these concerns reflect a broader distrust of the professional and regulatory entities responsible for the approval, use, and ongoing evaluation of new health interventions. Significantly, this distrust can

be rooted in complex public sentiments or troubled historical experiences and can play an important role in shaping public acceptance and patterns of use of any new health intervention.³¹ It is sobering, for example, that although none of the concerns about vaccine use have been supported by research, a significant number of parents still refuse or delay vaccinating their children.

In addition to these broad concerns, actual access to appropriate assistive technologies for disabled children depends heavily upon the health care and education systems, both of which are increasingly vulnerable to political pressure to reduce expenditures on public programs. Beyond this general financial pressure, however, lies a series of specific challenges within pediatrics and the child health care delivery system that must also be confronted if any real improvements in the quality of services provided to children with disabilities are to be made.

Assessing the Capacity of Current Delivery Systems

Any assessment of the delivery mechanisms for new technologies available for children with disabilities must begin with an examination of the capacity of the pediatric community to provide high-quality care for children with chronic conditions. In this respect, there is substantial reason for concern. Without important reforms, the current system of child health care in the United States will prove increasingly incapable of ensuring the dissemination and appropriate use of innovative technologies for children with serious disabling conditions.

Pediatric Capability for Comprehensive Care

Over the past several decades, the threat of serious, acute infection in young children has